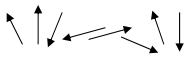
Inorganic experiment 2: Metallocene

Session II-b (ii) Measurement of magnetic susceptibility

- Difference between paramagnetism and diamagnetism

Types	*Value of magnetic susceptibility (χ)*		*Magnetic Behaviour*
Paramagnetism	small & positive (10^{-5} to 10^{-3})		unpaired electrons are randomly arranged
Diamagnetism	small & negative (-10^{-6} to -10^{-5})		No unpaired electron → No magnetic moment

- Other types of magnetism: Ferromagnetism, Antiferromagnetism Ferrimagnetism → covered in CM3111

like % yield, calculate deviation from expected, literature value. if not within -10^{-6} – 10^{-5} or 10^{3} – 10^{5}, discuss why in the report.

ch1 강기

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Effective Magnetic Moment: spin-only for 1st row *d*-block complexes

- For most of first row *d*-block metal complexes, the magnetic moment (μ) can be determined by the number of unpaired electrons (n).

Since total spin quantum number (S) = $\frac{n}{2}$,

$\mu_s = \mu_{eff}$(spin only) = μ(spin-only) = $2\sqrt{S(S+1)} = \sqrt{n(n+2)}$ (units: "Bohr Magnetons", B.M. or μ_B)

~ based on spin only ~
↳ okay for 1st row of transition metal

lower E transition metal if e⁻ are unpaired (?), less repulsion (due to

- Example: Fe^{2+} is d^6, and when surrounded by six weak field ligands (water molecules) it is expected to be high-spin with 4 unpaired electrons.

aqua ligand
will fig(?) ligand ??
high spin
(gap b/w e_g & t_{2g} is small. so prefer less # of paired e⁻ to minimise repulsion

$\mu_s = [n(n+2)]^{1/2}$ (where n = 4 for high-spin Fe^{2+})
→ μ_s = ~ $[4(4+2)]^{1/2}$
→ μ_s = 4.90 B.M. (for high-spin Fe^{2+})

d^6 e_g t_{2g}

Reference: C. E. Housecroft, A. G. Sharpe, *Inorganic Chemistry*, 3rd ed. Pearson.

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Experimental Magnetic Moment

- Molar susceptibility of the paramagnetic atom $\chi_a = \chi_{M(total;\ expt)} - \chi_{Di}$ (Σ of diamagnetic corrections)
- Experimental (Measured) magnetic moment $\mu_{eff} = 2.828\ (\chi_a \times T)^{1/2}$

related to T. temp은 datasheet에 제공

Anion	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$	Anion	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$	Anion	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$	Anion	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$
AsO_3^{3-}	–51	$C_5H_5^-$	–65	NCO^-	–23	$S_2O_3^{2-}$	–46
AsO_4^{3-}	–60	$C_6H_5COO^-$	–71	NCS^-	–31.0	$S_2O_8^{2-}$	–78
BF_4^-	–37	CO_3^{2-}	–28.0	O^{2-}	–12.0[a]	HSO_4^-	–35.0
BO_3^{3-}	–35	$C_2O_4^{2-}$	–34	OAc^-	–31.5	Se^{2-}	–48[b]
Br^-	–34.6	F^-	–9.1	OH^-	–12.0	SeO_3^{2-}	–44
BrO_3^-	–40	$HCOO^-$	–17	PO_3^{3-}	–42	SeO_4^{2-}	–51
Cl^-	–23.4	I^-	–50.6	$PtCl_6^{2-}$	–148	SiO_3^{2-}	–36
ClO_3^-	–30.2	IO_3^-	–51	S^{2-}	–30	Te^{2-}	–70
ClO_4^-	–32.0	IO_4^-	–51.9	SO_3^{2-}	–38	TeO_3^{2-}	–63
CN^-	–13.0	NO_2^-	–10.0	SO_4^{2-}	–40.1	TeO_4^{2-}	–55
		NO_3^-	–18.9				

Cation	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$
Fe^{2+}	–13
Fe^{3+}	–10

some magnetic correction for metal itself as their ave e⁻ in 4s, 3p etc.

1 erg/G = 1 emu → 10^{-3} A·m^2 = 10^{-3} J/T

Ligand	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$	Ligand	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$	Ligand	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$	Ligand	$\chi_{Di}/(1 \times 10^{-6}$ emu mol$^{-1})$
Acac$^-$	–52	Ethylene	–15	NH_3	–18	Pyrazine	–50
Bipy	–105	Glycinate	–37	Phen	–128	Pyridine	–49
CO	–10	H_2O	–13	o-PBMA	–194	Salen^{2-}	–182
$C_5H_5^-$	–65	Hyrdazine	–20	Phthalocyanine	–442	Urea	–34
En	–46.5	Malonate	–45	PPh_3	–167		

NOTE: Abbreviations: acac = acetylacetonate, bipy = 2,2′-dipyridyl, en = ethylenediamine, phen = phenanthroline, PBMA = phenylenebisdimethylarsine, salen = ethylenebis(salicylaminate)

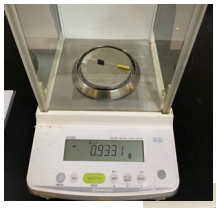
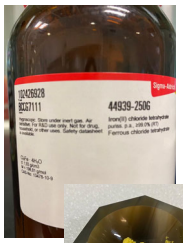
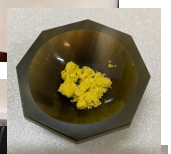
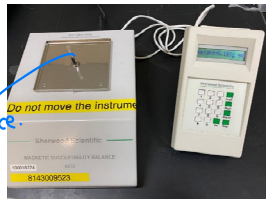
Reference: *J. Chem. Educ.* **2008**, 85, 4, 532

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Inorganic experiment 2: Metallocene

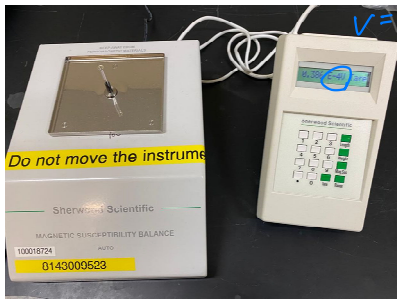
Session II-b (ii) Measurement of magnetic susceptibility

Example: **Iron(II) chloride tetrahydrate**

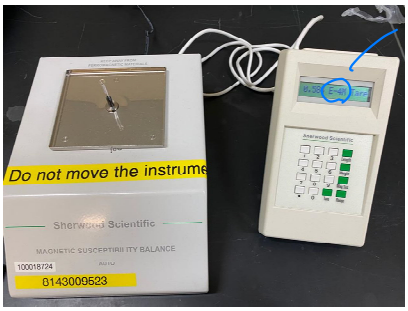


ensure no contact on the surface.





V = vol susceptibility value.



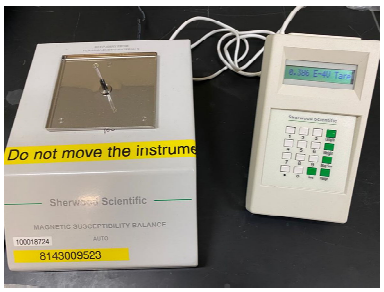


mass susceptibility.

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Inorganic experiment 2: Metallocene

Session II-b (ii) Measurement of magnetic susceptibility



- There are three magnetic susceptibilities, χ_v , χ_g , χ_M
- χ_v = volume susceptibility which is dimensionless → by experiment
- χ_g = mass susceptibility = gram susceptibility
  → by conversion
  χ_m 에서 계산하기도 함

$$\chi_g = \frac{\chi_v}{p} \quad \text{(units: } m^3kg^{-1}\text{)}$$ where p is the density of the sample

- χ_M = molar susceptibility → by calculation

$$\chi_M = \chi_g(MW) \quad \text{(SI units: } m^3mol^{-1}\text{)}$$ where MW is the molecular mass of the compound

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Inorganic experiment 2: Metallocene

- *Measurement of magnetic susceptibility for Iron(II) chloride tetrahydrate, $FeCl_2.4H_2O$*

$FeCl_2.4H_2O$ Mass susceptibility $\chi_g = 0.584 \times 10^{-4}$ erg G^{-1} $g^{-1} = 5.84 \times 10^{-5}$ erg G^{-1} g^{-1}

starting mat. compare this w product value

Temperature: 22 degC (i.e. 295 K)

Molar susceptibility χ_M (total) = χ_g (total) x M.W.

= $(5.84 \times 10^{-5}$ erg G^{-1} $g^{-1})(198.81$ g $mol^{-1})$

= 1.16×10^{-2} erg G^{-1} mol^{-1}

Correction for diamagnetic contributions (χ Di.) for 2 Cl^- and $4H_2O$:

$-(23.4 \times 2 + 13 \times 4) \times 10^{-6}$ erg G^{-1} mol^{-1}

= -0.99×10^{-4} erg G^{-1} mol^{-1}

If the metal centre is considered as purely cationic, then diamagnetic contributions for Fe^{2+} is considered

→ χ Di. = $-0.99 \times 10^{-4} + (-13 \times 10^{-6})$ erg G^{-1} mol^{-1} = -1.12×10^{-4} erg G^{-1} mol^{-1}

χ_M total = χ para + χ Di. (Ligand + ions)

Therefore

$\chi_a = \chi$ para = χ corr. = χ_M total - χ di.

covalent
if consider, assuming no interxn b/w metal & ligand. only ionic.
실제. some lit value x consider Fe^{2+} contribution

molar susceptibility of the paramagnetic atom, χ para = χ_M total - Σ of diamagnetic corrections (χ Di.)

$\chi_a = \chi$ para = 1.16×10^{-2} erg G^{-2} $mol^{-1} - (-0.99 \times 10^{-4})$ erg G^{-1} $mol^{-1} = 1.17 \times 10^{-2}$ erg G^{-1} mol^{-1}

→ $\chi_{para} = 1.16 \times 10^{-2}$ erg G^{-1} $mol^{-1} - (-1.12 \times 10^{-4})$ erg G^{-1} $mol^{-1} = -1.17 \times 10^{-2}$ erg G^{-1} mol^{-1}

Experimental (Measured) magnetic moment $\mu_{eff} = 2.828\ (\chi_a \times T)^{1/2}$

= $2.828\ [(1.17 \times 10^{-2}$ erg G^{-1} $mol^{-1})(295$ K$)]^{½}$

= **5.25 B.M.** (→ ~7 % higher than the calculated value μ_s) → Find **n** by using $\mu_s = [n(n+2)]^{1/2}$

Compare this magnetic susceptibility of $FeCl_2.4H_2O$ and the magnetic susceptibility measured for your purified ferrocene obtained!!

Thus, the measured effective magnetic moment (μ_{eff}) is about 7% higher than the predicted spin-only value of magnetic moment , but the result is consistent with a high-spin Fe^{2+} (with 4 unpaired electrons).

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Supplementary

- *Measurement of magnetic susceptibility for Iron(II) chloride tetrahydrate, $FeCl_2.4H_2O$*

Remarks:

- The real materials/compounds often exist between the two extremes of purely covalent and purely ionic environments.
- If the metal centre of the mononuclear complex is considered as purely cationic (i.e. without covalent interaction with ligands; refer to *Communications Physics*, **2022**, 5, 95, and *J. Chem. Educ.*, **2008**, 85, 532), the diamagnetic correction (χ_{Di}) for such metal cation, which is also stated as the "diamagnetic correction for inner-core electrons due to transition metal" generally in some literatures, should be taken into consideration in the calculation of molar susceptibility of paramagnetic atom. If it is applied in iron(II) chloride tetrahydrate, the outcomes remain the same in the calculation given as shown on p.172.
- The above estimation may not be very accurate when the bond between metal and ligand has covalent character. The most accurate χ_{Di} values can only be obtained by measurement.
- Hints: Both chloride and aqua (water) ligands are σ and π donors although aqua (water) is weaker π-donor ligand. Cyclopentadienyl ligand has the common η^5-bonding mode due to the presence of both σ- and π- donation in conjunction with δ(back)bonding feature. In some literatures, cyclopentadienyl ligand was stated to have π(back)bonding character.

References: (i) *Communications Physics*, **2022**, 5, 95; (ii) *J. Chem. Educ.*, **2008**, 85, 532.

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Session II-b (iii) UV-visible Spectroscopic Characterization of the purified ferrocene

UV-Visible Spectroscopy

Beer-Lambert Law / Beer's Law

The *intensity* of the absorption determines whether the colour will be pale or dark. Beer's Law relates the absorbance of a peak to the molar absorptivity or extinction coefficient, ε, of each band in the UV-visible spectrum (*i.e.*, each absorbance maximum, λ_{max}, has an associated value of ε):

$A = \varepsilon \ell c$

where A = absorbance (vertical scale on the spectrum)
ε = extinction coefficient (in $cm^{-1} \cdot M^{-1}$)
ℓ = the path length of the cell (in cm)
c = the concentration of the solution (in M)

$$A = -\log_{10}\left(\frac{I}{I_o}\right)$$

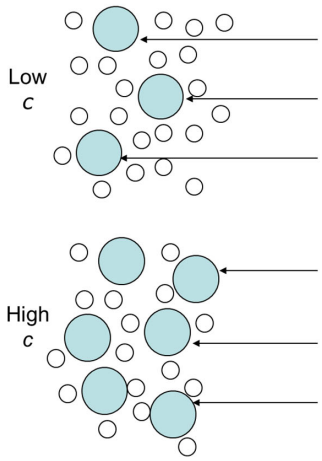
where
I_o = intensity of incident light;
I = intensity of light leaving sample

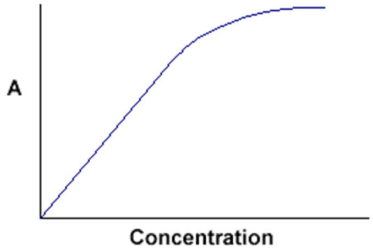
Source: J. Cooke @ Univ. of Alberta

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UV-Visible Spectroscopy

- Deviations from Beer-Lambert Law: sample concentration





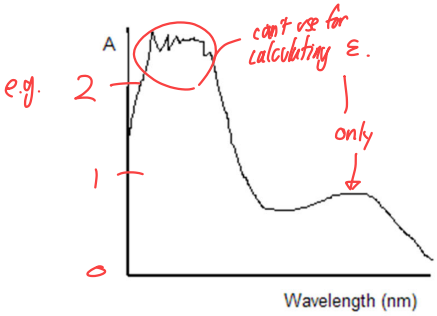
The Beer-Lambert Law assumes that all molecules contribute to the absorption and that no absorbing molecule is in the shadow of another.

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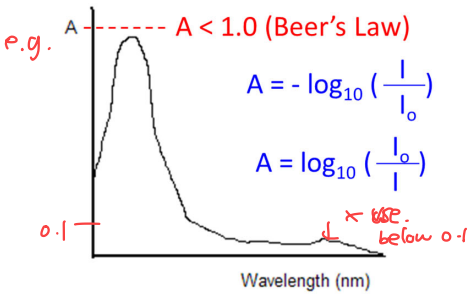
UV-Visible Spectroscopy

- Deviations from ~~Beer-Lambert Law~~: sample concentration

A must be 0.1 – 1.0



Solution is too concentrated



Diluted five-fold

example only

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Electron Counting with CBC method

This topic is related to "Experiment 1: Homogenous Catalysis & Experiment 2: Metallocene".

Disclaimer: All materials, information, weblinks and pictures contained in this video are for educational purposes only.

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Electron Counting for the complexes containing metal-ligand covalent interaction

since there can also be covalent character b/w M-L

- Donor-pair method of electron counting is misleading in organometallic chemistry and for the complex containing metal-ligand covalent interaction:

  (i) a recent IUPAC article concludes that it is inappropriate to assign oxidation numbers with respect to the nomenclature of organometallic compounds

  (ii) the assignment of "coordination number" is often ambiguous for organometallic compounds because the term is interpreted with more than one meaning in the literature.

- Prof. Malcolm Green introduced an innovative method for the formal classification of covalent compounds, "Covalent Bond Classification (CBC) Method" and published it in 1995 in Journal of Organometallic Chemistry.
- The main advantage of the "Covalent Bond Classification (CBC) Method" is that it was specifically designed for covalent molecules.
- CBC method is more relevant to classify and provide insight into the nature of *covalent* organometallic molecules.
- It is also called neutral ligand method of electron counting.

Sources:
(i) *J. Organomet. Chem.* 1995, *500*, 127-148;
(ii) http://www.columbia.edu/cu/chemistry/groups/parkin/cbc.htm

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Covalent Bond Classification (CBC) Method

The CBC method is based on the notion that there are three basic types of interaction by which a ligand may bond to a metal center and the ligand is classified according to the nature and number of these interactions. The three basic types of interaction are represented by the symbols L, X, and Z, which correspond respectively to 2-electron, 1-electron and 0-electron *neutral* ligands and are clearly differentiated according to a molecular orbital representation of the bonding.

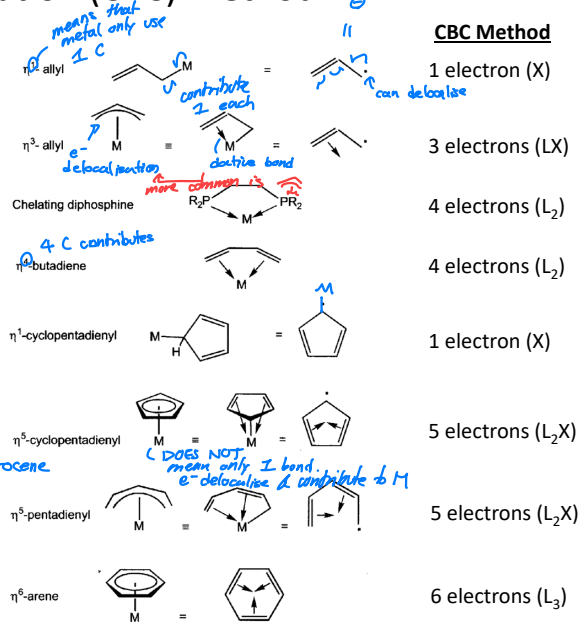
- ***An L-function ligand*** donates two electrons to a metal center *via* a dative covalent bond. This type of ligand contribute 2 electrons in electron counting. Simple examples of L-type ligands include R_3P, R_2O, and CO, *i.e.* donor molecules that have lone pairs (Lewis bases).
- ***An X-function ligand*** is one which interacts with a metal center *via* a normal 2-electron covalent bond, composed of 1 electron from the metal and 1 electron from the X ligand. This type of ligand contribute 1 electron in electron counting. Simple examples of X-type ligands include H, X (X = halide) and CH_3, *i.e.* radicals.
- ***A Z-function ligand*** also interacts with a metal center *via* a dative covalent bond, but differs from the L-function in that both electrons are donated by the metal rather than the ligand. As such, a Z-function ligand donates zero electrons to a metal center. Simple examples of Z-type ligands include BF_3, BR_3 and AlR_3, *i.e.* molecules that have a vacant orbital (Lewis acids).

Sources:
(i) *J. Organomet. Chem.* 1995, *500*, 127-148;
(ii) http://www.columbia.edu/cu/chemistry/groups/parkin/cbc.htm

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Covalent Bond Classification (CBC) Method

- Examples:



CBC Method
1 electron (X)
3 electrons (LX)
4 electrons (L_2)
4 electrons (L_2)
1 electron (X)
5 electrons (L_2X)
5 electrons (L_2X)
6 electrons (L_3)

Source: http://www.lifesci.sussex.ac.uk

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Covalent Bond Classification (CBC) Method

- **Oxidation State (O.S.)**

  The number of X-function ligands in neutral compounds is equal to the oxidation state of the metal.

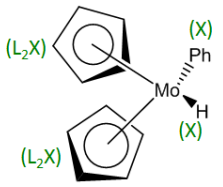
  L : dative bond in Pt

  Examples: $[Mn(CO)_5Me] = MnL_5X \rightarrow$ O.S. = 1
  $[Pt(CO)_2Cl_2] = Pt\,L_2X_2 \rightarrow$ O.S. = 2

- **Coordination Number (C.N.)**

  It is equal to the total numbers of L-function and X-function ligands, instead of the number of donor atoms of the ligands coordinated to the metal.

  Example:



Source: http://www.lifesci.sussex.ac.uk

$[Mo(L_2X)_2(X)(X)] = MoL_4X_4 \rightarrow$ C.N.= 8, O.S. = 4

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Covalent Bond Classification (CBC) Method

- **Anionic Complexes**

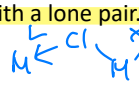
  In an anion, one is added to the electron count for each negative charge. For its O.S., one is subtracted from the number of Xs for each negative charge.

- **Cationic Complexes**

  In a cation, one is subtracted from the electron count for each positive charge. For its O.S., one is added to the number of Xs for each positive charge.

- **Bridging Ligands**

  Halido ligands can act as bridging ligands between two metal centres with a lone pair. They count as LX in which X is for sigma bond while L is for dative bond.



↳ LX instead of X

Sources:
(i) *J. Organomet. Chem.* 1995, *500*, 127-148;
(ii) http://www.columbia.edu/cu/chemistry/groups/parkin/cbc.htm

CH4

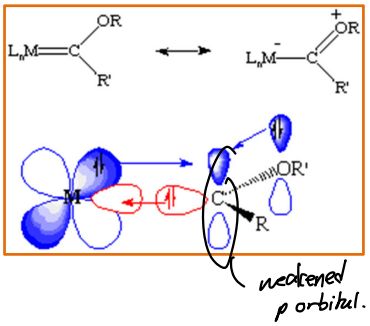
Covalent Bond Classification (CBC) Method

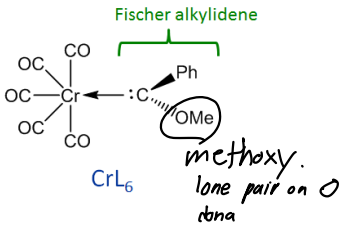
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- **Alkylidene and Alkylidyne ligands**

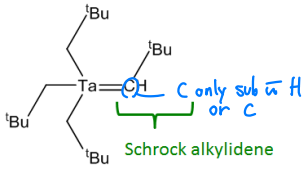
There are two classes of alkylidenes: → The details are covered in CM3111.
(i) Fischer alkylidenes: alkylidene carbon is electrophilic, which are classified as L ligands;
(ii) Schrock alkylidenes: alkylidene carbon is nucleophilic, which are classified as X_2 ligands.

Similarly, Fischer alkylidynes are classified as LX and Schrock alkylidynes as X_3.





CrL_6



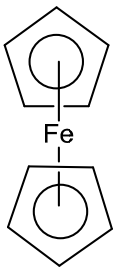
TaX_5

Source of picture: http://www.ilpi.com/organomet/carbene.html

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Exercise 1

Using CBC method, deduce the number of valence electrons, oxidation state and coordination number of the metal centre for the following complex. Detailed calculation steps must be shown.



$[Fe(C_5H_5)_2] \rightarrow FeL_4X_2 \rightarrow$ O.S. = 2, C.N. = 6

considering all atoms to be neutral,

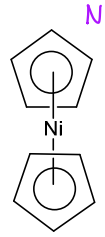
Fe 8 e⁻
2 C_p 10 e⁻
18 e⁻ of valence e⁻

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Exercise 2

- Using **CBC method**, deduce the number of valence electrons, oxidation state and coordination number of metal centre for the following complexes. Detailed calculation steps must be shown.

(i)



NiL_4X_2

∴ OS: +2

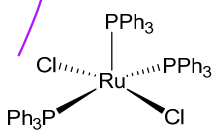
CN: 6

Ni: 10

+4x2 +2

= 20 valence e⁻

(ii)



RuL_3X_2

∴ OS: +2

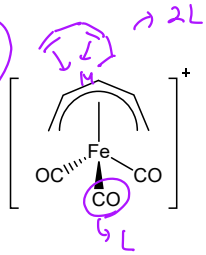
CN: 5

Ru: 8

+ 3x2+2

= 16 valence e⁻

(iii)



2L & 1X

$[FeL_5X]^+$

∴ OS: +2

CN: 6

valence e⁻

Fe: 8

5x2 +1 −1

= 18

Reminder

Experiment-Based Risk Assessment Form

Name of Department _____ Name and Location of Lab _____

Name of Laboratory _____ Name of PI (lecturer-in-charge) _____

Name of Student _____ Name of Activity/Experiment _____

	Hazard Identification			Risk Evaluation & Control						
No	Description / Details of Steps in Activity	Hazard(s)	Possible Accident(s)	Existing Risk Control (Mitigation)	Severity	Likelihood (probability)	Risk Level	Additional Risk Control	Person Responsible	By (Date)

Conducted By _____ Approved By _____

Name _____ Name _____

Signature _____ Signature _____

date _____ Approval date _____ Next Revision date (Maximum 3 years) _____

- **Submit your filled risk assessment for inorganic expt. 2 (session II) *by* 10:00 a.m. on 16-Sep-2024 (Mon.; Week 6) via Canvas-Assignments [5% CA]. Penalty will be applied for any late submission.**